Physics : Section-A (Q. No. 1 to 35)

1 Let a wire be suspended from the ceiling (rigid support) and stretched by a weight W attached at its free end. The longitudinal stress at any point of cross-sectional area A of the wire is :

(1) $2W/A$ (2) W/A

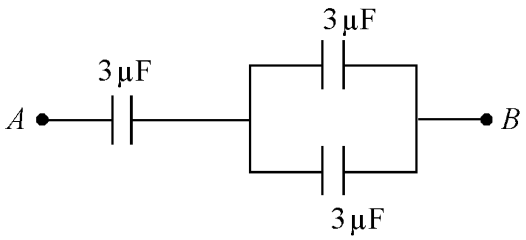
(3) $W/2A$ (4) Zero

2 The ratio of radius of gyration of a solid sphere of mass M and radius R about its own axis to the radius of gyration of the thin hollow sphere of same mass and radius about its axis is :

(1) 3 : 5 (2) 5 : 3

(3) 2 : 5 (4) 5 : 2

3 The equivalent capacitance of the system shown in the following circuit is :



(1) 2 μF (2) 3 μF

(3) 6 μF (4) 9 μF

4 A football player is moving southward and suddenly turns eastward with the same speed to avoid an opponent. The force that acts on the player while turning is :

(1) along eastward

(2) along northward

(3) along north-east

(4) along south-west

5 If $\oint_s \vec{E} \cdot \vec{dS} = 0$ over a surface, then :

(1) the number of flux lines entering the surface must be equal to the number of flux lines leaving it.

(2) the magnitude of electric field on the surface is constant.

(3) all the charges must necessarily be inside the surface.

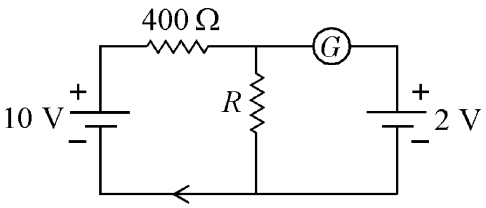
(4) the electric field inside the surface is necessarily uniform.

6 The potential energy of a long spring when stretched by 2 cm is U. If the spring is stretched by 8 cm, potential energy stored in it will be :

(1) 2U (2) 4U

(3) 8U (4) 16U

7 If the galvanometer G does not show any deflection in the circuit shown, the value of R is given by :



(1) 200 Ω (2) 50 Ω

(3) 100 Ω (4) 400 Ω

8 A 12 V, 60 W lamp is connected to the secondary of a step down transformer, whose primary is connected to ac mains of 220 V. Assuming the transformer to be ideal, what is the current in the primary winding?

(1) 0.27 A (2) 2.7 A

(3) 3.7 A (4) 0.37 A

9 A full wave rectifier circuit consists of two p-n junction diodes, a centre-tapped transformer, capacitor and a load resistance. Which of these components remove the ac ripple from the rectified output?

(1) A centre-tapped transformer
(2) p-n junction diodes
(3) Capacitor
(4) Load resistance

10 Light travels a distance x in time t_1 in air and $10x$ in time t_2 in another denser medium. What is the critical angle for this medium?

(1) $\sin^{-1}\left(\frac{t_2}{t_1}\right)$ (2) $\sin^{-1}\left(\frac{10\,t_2}{t_1}\right)$

(3) $\sin^{-1}\left(\frac{t_1}{10\,t_2}\right)$ (4) $\sin^{-1}\left(\frac{10\,t_1}{t_2}\right)$

11 Resistance of a carbon resistor determined from colour codes is $(22000 \pm 5\%)\,\Omega$. The colour of third band must be :

(1) Red (2) Green
(3) Orange (4) Yellow

12 Given below are two statements:

Statement I : Photovoltaic devices can convert optical radiation into electricity.

Statement II : Zener diode is designed to operate under reverse bias in breakdown region.

In the light of the above statements, choose the ***most appropriate*** answer from the options given below :

(1) Both **Statement I** and **Statement II** are correct.
(2) Both **Statement I** and **Statement II** are incorrect.
(3) **Statement I** is correct but **Statement II** is incorrect.
(4) **Statement I** is incorrect but **Statement II** is correct.

13 The magnetic energy stored in an inductor of inductance $4\ \mu H$ carrying a current of 2 A is :

(1) $4\ \mu J$ (2) $4\ mJ$
(3) $8\ mJ$ (4) $8\ \mu J$

14 The angular acceleration of a body, moving along the circumference of a circle, is :

(1) along the radius, away from centre
(2) along the radius towards the centre
(3) along the tangent to its position
(4) along the axis of rotation

15 A Carnot engine has an efficiency of 50% when its source is at a temperature 327° C. The temperature of the sink is :

(1) 27° C (2) 15° C
(3) 100° C (4) 200° C

16 Two bodies of mass m and $9m$ are placed at a distance R. The gravitational potential on the line joining the bodies where the gravitational field equals zero, will be (G = gravitational constant) :

(1) $-\frac{8\,Gm}{R}$ (2) $-\frac{12\,Gm}{R}$

(3) $-\frac{16\,Gm}{R}$ (4) $-\frac{20\,Gm}{R}$

17 A vehicle travels half the distance with speed ϑ and the remaining distance with speed 2ϑ. Its average speed is:

(1) $\frac{\vartheta}{3}$ (2) $\frac{2\vartheta}{3}$

(3) $\frac{4\vartheta}{3}$ (4) $\frac{3\vartheta}{4}$

18 The amount of energy required to form a soap bubble of radius 2 cm from a soap solution is nearly : (surface tension of soap solution = 0.03 N m^{-1})

(1) 30.16×10^{-4}J (2) 5.06×10^{-4}J

(3) 3.01×10^{-4}J (4) 50.1×10^{-4}J

19 The minimum wavelength of X-rays produced by an electron accelerated through a potential difference of V volts is proportional to:

(1) $\sqrt{V}$ (2) $\frac{1}{V}$

(3) $\frac{1}{\sqrt{V}}$ (4) V^2

20 The half life of a radioactive substance is 20 minutes. In how much time, the activity of substance drops to $\left(\frac{1}{16}\right)^{th}$ of its initial value?

(1) 20 minutes (2) 40 minutes

(3) 60 minutes (4) 80 minutes

21 A metal wire has mass (0.4 ± 0.002) g, radius (0.3 ± 0.001) mm and length (5 ± 0.02) cm. The maximum possible percentage error in the measurement of density will nearly be:

(1) 1.2% (2) 1.3%

(3) 1.6% (4) 1.4%

22 In a plane electromagnetic wave travelling in free space, the electric field component oscillates sinusoidally at a frequency of 2.0×10^{10}Hz and amplitude $48\,V\,m^{-1}$. Then the amplitude of oscillating magnetic field is : (Speed of light in free space = 3×10^8 m s^{-1})

(1) 1.6×10^{-9}T (2) 1.6×10^{-8}T

(3) 1.6×10^{-7}T (4) 1.6×10^{-6}T

23 The temperature of a gas is –50° C. To what temperature the gas should be heated so that the rms speed is increased by 3 times?

(1) 669° C (2) 3295° C

(3) 3097 K (4) 223 K

24 An ac source is connected to a capacitor C. Due to decrease in its operating frequency :

(1) capacitive reactance decreases.

(2) displacement current increases.

(3) displacement current decreases.

(4) capacitive reactance remains constant

25 For Young's double slit experiment, two statements are given below:

Statement I : If screen is moved away from the plane of slits, angular separation of the fringes remains constant.

Statement II : If the monochromatic source is replaced by another monochromatic source of larger wavelength, the angular separation of fringes decreases.

In the light of the above statements, choose the *correct* answer from the options given below:

(1) Both **Statement I** and **Statement II** are true.

(2) Both **Statement I** and **Statement II** are false.

(3) **Statement I** is true but **Statement II** is false.

(4) **Statement I** is false but **Statement II** is true.

26 In hydrogen spectrum, the shortest wavelength in the Balmer series is λ. The shortest wavelength in the Bracket series is :

(1) 2λ (2) 4λ

(3) 9λ (4) 16λ

27 The work functions of Caesium (Cs), Potassium (K) and Sodium (Na) are 2.14 eV, 2.30 eV and 2.75 eV respectively. If incident electromagnetic radiation has an incident energy of 2.20 eV, which of these photosensitive surfaces may emit photoelectrons?

(1) Cs only
(2) Both Na and K
(3) K only
(4) Na only

28 The errors in the measurement which arise due to unpredictable fluctuations in temperature and voltage supply are :

(1) Instrumental errors
(2) Personal errors
(3) Least count errors
(4) Random errors

29 In a series *LCR* circuit, the inductance L is 10 mH, capacitance C is $1\ \mu F$ and resistance R is $100\ \Omega$. The frequency at which resonance occurs is :

(1) 15.9 rad/s (2) 15.9 kHz
(3) 1.59 rad/s (4) 1.59 kHz

30 The venturi-meter works on :

(1) Huygen's principle
(2) Bernoulli's principle
(3) The principle of parallel axes
(4) The principle of perpendicular axes

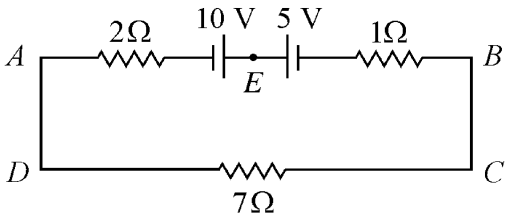
31 The ratio of frequencies of fundamental harmonic produced by an open pipe to that of closed pipe having the same length is :

(1) 1 : 2 (2) 2 : 1
(3) 1 : 3 (4) 3 : 1

32 An electric dipole is placed at an angle of 30° with an electric field of intensity $2\times10^5\ N\,C^{-1}$. It experiences a torque equal to 4 N m. Calculate the magnitude of charge on the dipole, if the dipole length is 2 cm.

(1) 8 mC (2) 6 mC
(3) 4 mC (4) 2 mC

33 The magnitude and direction of the current in the following circuit is



(1) 0.2 A from B to A through E
(2) 0.5 A from A to B through E
(3) $\frac{5}{9}$ A from A to B through E
(4) 1.5 A from B to A through E

34 The net magnetic flux through any closed surface is :

(1) Zero (2) Positive
(3) Infinity (4) Negative

35 A bullet is fired from a gun at the speed of $280\ m\ s^{-1}$ in the direction 30° above the horizontal. The maximum height attained by the bullet is $\left(g = 9.8\ m\ s^{-2},\ \sin 30° = 0.5\right)$:

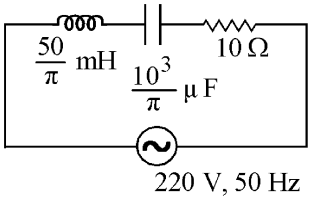
(1) 2800 m (2) 2000 m
(3) 1000 m (4) 3000 m

36 Two thin lenses are of same focal lengths (f), but one is convex and the other one is concave. When they are placed in contact with each other, the equivalent focal length of the combination will be :

(1) Zero (2) $f/4$

(3) $f/2$ (4) Infinite

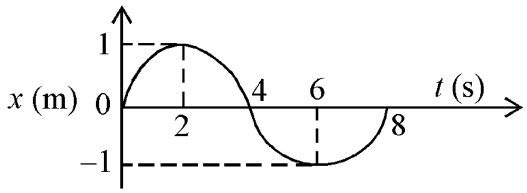
37 The net impedance of circuit (as shown in figure) will be :



(1) $10\sqrt{2}\ \Omega$ (2) $15\ \Omega$

(3) $5\sqrt{5}\ \Omega$ (4) $25\ \Omega$

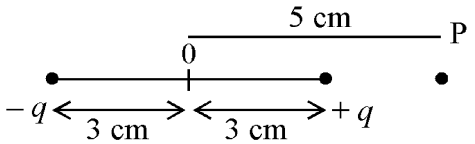
38 The x-t graph of a particle performing simple harmonic motion is shown in the figure. The acceleration of the particle at $t = 2$ s is :



(1) $\frac{\pi^2}{8}\ m\,s^{-2}$ (2) $-\frac{\pi^2}{8}\ m\,s^{-2}$

(3) $\frac{\pi^2}{16}\ m\,s^{-2}$ (4) $-\frac{\pi^2}{16}\ m\,s^{-2}$

39 An electric dipole is placed as shown in the figure.



The electric potential (in 10^2 V) at point P due to the dipole is ($\in_0$ = permittivity of free space and $\frac{1}{4\pi\in_0} = K$):

(1) $\left(\frac{3}{8}\right)qK$ (2) $\left(\frac{5}{8}\right)qK$

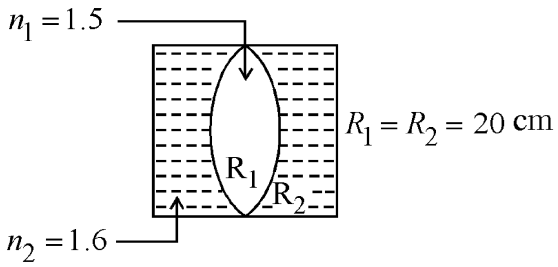
(3) $\left(\frac{8}{5}\right)qK$ (4) $\left(\frac{8}{3}\right)qK$

40 A bullet from a gun is fired on a rectangular wooden block with velocity u. When bullet travels 24 cm through the block along its length horizontally, velocity of bullet becomes $\frac{u}{3}$. Then it further penetrates into the block in the same direction before coming to rest exactly at the other end of the block. The total length of the block is :

(1) 27 cm (2) 24 cm

(3) 28 cm (4) 30 cm

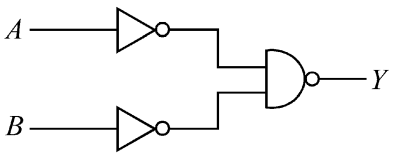
41 In the figure shown here, what is the equivalent focal length of the combination of lenses (Assume that all layers are thin)?



(1) 40 cm (2) − 40 cm

(3) − 100 cm (4) − 50 cm

42 For the following logic circuit, the truth table is:



(1)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(3)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(4)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

43 A horizontal bridge is built across a river. A student standing on the bridge throws a small ball vertically upwards with a velocity $4\ \text{m s}^{-1}$. The ball strikes the water surface after $4\ \text{s}$. The height of bridge above water surface is (Take $g = 10\ \text{m s}^{-2}$) :

(1) 56 m (2) 60 m

(3) 64 m (4) 68 m

44 10 resistors, each of resistance R are connected in series to a battery of emf E and negligible internal resistance. Then those are connected in parallel to the same battery, the current is increased n times. The value of n is :

(1) 10 (2) 100

(3) 1 (4) 1000

45 A wire carrying a current I along the positive x-axis has length L. It is kept in a magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} - 4\hat{k})$ T. The magnitude of the magnetic force acting on the wire is :

(1) $3\ IL$ (2) $\sqrt{5}\ IL$

(3) $5\ IL$ (4) $\sqrt{3}\ IL$

46 A satellite is orbiting just above the surface of the earth with period T. If d is the density of the earth and G is the universal constant of gravitation, the quantity $\frac{3\pi}{Gd}$ represents :

(1) T (2) T^2

(3) T^3 (4) $\sqrt{T}$

47 Calculate the maximum acceleration of a moving car so that a body lying on the floor of the car remains stationary. The coefficient of static friction between the body and the floor is 0.15 ($g = 10\ \text{m s}^{-2}$).

(1) $1.2\ \text{m s}^{-2}$ (2) $150\ \text{m s}^{-2}$

(3) $1.5\ \text{m s}^{-2}$ (4) $50\ \text{m s}^{-2}$

48 The resistance of platinum wire at 0°C is 2Ω and 6.8Ω at 80°C. The temperature coefficient of resistance of the wire is :

(1) $3\times10^{-4}\ °\text{C}^{-1}$ (2) $3\times10^{-3}\ °\text{C}^{-1}$

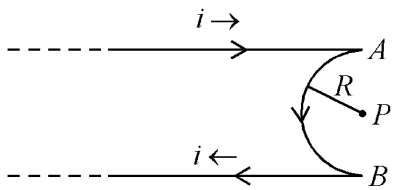
(3) $3\times10^{-2}\ °\text{C}^{-1}$ (4) $3\times10^{-1}\ °\text{C}^{-1}$

49 The radius of inner most orbit of hydrogen atom is $5.3\times10^{-11}\ \text{m}$. What is the radius of third allowed orbit of hydrogen atom?

(1) $0.53\ \text{Å}$ (2) $1.06\ \text{Å}$

(3) $1.59\ \text{Å}$ (4) $4.77\ \text{Å}$

50 A very long conducting wire is bent in a semi-circular shape from A to B as shown in figure. The magnetic field at point P for steady current configuration is given by :



(1) $\frac{\mu_0 i}{4R}$ pointed into the page

(2) $\frac{\mu_0 i}{4R}$ pointed away from the page

(3) $\frac{\mu_0 i}{4R}\left[1-\frac{2}{\pi}\right]$ pointed away from page

(4) $\frac{\mu_0 i}{4R}\left[1-\frac{2}{\pi}\right]$ pointed into the page

51 Which of the following reactions will NOT give primary amine as the product?

(1) $CH_3\,CONH_2 \xrightarrow{Br_2\,/\,KOH}$ Product

(2) $CH_3CN \xrightarrow[(ii)\,H_3O\oplus]{(i)\,LiAlH_4}$ Product

(3) $CH_3NC \xrightarrow[(ii)\,H_3O\oplus]{(i)\,LiAlH_4}$ Product

(4) $CH_3CONH_2 \xrightarrow[(ii)\,H_3O\oplus]{(i)\,LiAlH_4}$ Product

52 Match **List - I** with **List - II** :

	List - I		**List - II**
A.	Coke	I.	Carbon atoms are sp^3 hybridised.
B.	Diamond	II.	Used as a dry lubricant
C.	Fullerene	III.	Used as a reducing agent
D.	Graphite	IV.	Cage like molecules

Choose the **correct** answer from the options given below :

(1) A-II, B-IV, C-I, D-III
(2) A-IV, B-I, C-II, D-III
(3) A-III, B-I, C-IV, D-II
(4) A-III, B-IV, C-I, D-II

53 Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R** :

Assertion A : Metallic sodium dissolves in liquid ammonia giving a deep blue solution, which is paramagnetic.

Reasons R : The deep blue solution is due to the formation of amide.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(2) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**.
(3) **A** is true but **R** is false.
(4) **A** is false but **R** is true.

54 In Lassaigne's extract of an organic compound, both nitrogen and sulphur are present, which gives blood red colour with Fe^{3+} due to the formation of -

(1) $Fe_4\left[Fe(CN)_6\right]_3 \cdot x\,H_2O$

(2) NaSCN

(3) $\left[Fe(CN)_5\,NOS\right]^{4-}$

(4) $\left[Fe(SCN)\right]^{2+}$

55 The conductivity of centimolar solution of KCl at 25°C is 0.0210 $ohm^{-1}\,cm^{-1}$ and the resistance of the cell containing the solution at 25°C is 60 ohm. The value of cell constant is -

(1) 1.34 cm^{-1} (2) 3.28 cm^{-1}
(3) 1.26 cm^{-1} (4) 3.34 cm^{-1}

56 Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R** :

Assertion A : A reaction can have zero activation energy.

Reasons R : The minimum extra amount of energy absorbed by reactant molecules so that their energy becomes equal to threshold value, is called activation energy.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(2) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**.
(3) **A** is true but **R** is false.
(4) **A** is false but **R** is true.